KSP Datathon: The Master Blueprint

Your Best Friend

June 25, 2026

The Ultimate Step-by-Step Execution Plan

Step 1: Generate Synthetic Cybercrime Data

Goal: Create 1.5 Lakh rows of fake digital crime data.
Tools to Use: Python, Pandas, and the `Faker` library.

AI Prompt to Generate the Code

"Act as a Python Data Engineer. Write a highly optimized script using the 'Faker' library and Pandas to generate a synthetic dataset of 150,000 cybercrime records. The columns should include: Case_ID, Crypto_Wallet_Address, Dark_Web_URL, Suspect_IP, Transaction_Amount, and Timestamp. Ensure the code saves the output efficiently as a CSV file."

Step 2: Build the Agentic AI Core

Goal: Create an AI that can think, query the database, and link cases.
Tools to Use: Python, LangChain (framework), HuggingFace (for Open-Source LLMs like Llama-3 or Mistral).

AI Prompt to Generate the Code

"Act as an AI Architect. Write a Python script using LangChain to create an 'SQL Agent'. The agent must connect to a local SQLite/CSV database containing cybercrime data. Use an open-source LLM via HuggingFace endpoint. The agent should be able to receive a natural language query, convert it to SQL, execute it, and return a reasoned summary of the findings."

Step 3: Integrate Kannada Voice & Text (Bhashini)

Goal: Enable the chatbot to understand Kanglish and spoken Kannada.
Tools to Use: Bhashini API (National Language Translation Mission) or OpenAI Whisper for Speech-to-Text.

AI Prompt to Generate the Code

"Act as an NLP Engineer. Provide a Python integration script for the Bhashini API. The script should have two functions: 1) Convert spoken Kannada audio into Kannada text. 2) Translate transliterated 'Kanglish' text into formal English so it can be processed by an LLM."

Step 4: Deploy on Zoho Catalyst

Goal: Host your AI and Database securely on the mandatory cloud platform.
Tools to Use: Zoho Catalyst Relational Data Store (for the CSV), Docker (for containerization), and Zoho Catalyst AppSail (for hosting the Python app).

AI Prompt to Generate the Code

"Act as a DevOps Engineer. I need to deploy a Python FastAPI LangChain application on Zoho Catalyst AppSail. Write a complete 'Dockerfile' that installs all necessary dependencies (including LangChain and Pandas). Also, provide the deployment configuration steps required to push this Docker image to the Zoho Catalyst AppSail environment."

(Detailed Explanation in Kannada)

SHRIDHAR, :

1: (Data Generation)

, (Schema) . , (Dark Web), (Cryptocurrency) IP 1.5 .

- : (Python) (Faker) .
- ? CSV . AI .

2: (Agentic AI Core)

. (Bot) . " ?" , AI (Logical) .

- : LangChain (AI) Llama-3 - (Open-source) .
- ? LangChain 'SQL Agent' . SQL .

3: (Multilingual & Voice)

(Kanglish - "Kalla elli") , .

- : Bhashini API ( ) OpenAI Whisper.
- ? (Kanglish) Bhashini API , AI . AI .

4: (Zoho Catalyst Deployment)

, , Zoho .

- : Docker, Zoho Relational Data Store, Catalyst AppSail.
- ? 1.5 Zoho Data Store . 'Docker Image' Zoho AppSail (Host) . AI .

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